Many Alternatives, Random Incentives: Sharp Bounds and Efficient Estimation in Treatment Selection Models

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Instrumental Variables (IV) with Multiple Treatments

Observables: $(Y, D, Z) \sim P$

- Discrete "instrument" $Z \in \mathcal{Z}$: randomly assigned,
- Discrete "treatment" $D \in \mathcal{D}$: generally endogenous,
- Outcome(s) $Y \in \mathcal{Y}$ may be continuous: $\mathcal{Y}$ compact subset of $\mathbb{R}^k$.

Unobserved Heterogeneity: $\left(\left(Y_{d_1}, \dots, Y_{d_t}\right), \boldsymbol{T}, Z\right) \sim Q_0$

- Potential outcomes $Y_{d_1}, \dots, Y_{d_t}$ such that $Y = Y_D$,
- Response-type $\boldsymbol{T} := \left(D_{z_1}, \dots, D_{z_m}\right)$ such that $D = D_Z$.

Scope: Models that restrict how individuals select into treatment in response to instruments/incentives/assignment

This Paper: Unified (partial) identification approach

- Flexible choice of both target parameter and model
- **Sharp for "full" independence**, not just mean independence
- Estimation and Inference **without discretization**

Model Scope: Treatment Selection Models (TSMs)

Introduce unobservables $(\boldsymbol{V}, \varepsilon_Y)$, functions g_D and g_Y:

$$\text{Treatment Selection Equation: } D = g_D(Z, \boldsymbol{V}) \,,$$
$$\text{Outcome Equation: } Y = g_Y(D, \boldsymbol{V}, \varepsilon_Y) \,,$$
$$\text{Independence Condition: } Z, \boldsymbol{V}, \varepsilon_Y \text{ are mutually independent.}$$

Formally equivalent to the PO model, can define

$$Y_d := g_Y(d, \boldsymbol{V}, \varepsilon_Y) \quad \text{and} \quad D_z := g_D(z, \boldsymbol{V})$$
$$\rightarrow \ \boldsymbol{T} = \left(g_D(z, \boldsymbol{V}) : z \in \mathcal{Z}\right).$$

Treatment Selection Models: Restrictions on g_D or $\boldsymbol{V}$

- Examples: Latent index models, $D = \mathbb{1}\{\boldsymbol{V} \leq \nu(Z)\}$; encouragement designs; multiple threshold models; various types of instrument "monotonicity" (Unordered-, Partial-, Vector-Monotonicity)
- Applies naturally to sample selection & non-response bias models

→ Such models are characterized by restrictions on *distribution* of $\boldsymbol{T}$

Overview of Results

Identification

1. Sharp testable implications of such models,
   - Finite-dimensional convex representation (even with cts. Y),
   → **Sharp Test** for TSMs and IV Validity as **moment inequalities**,
2. Partial identification of $\mathbb{E}\left[Y_{d_1} - Y_{d_2} \,\middle|\, \boldsymbol{V} \in \mathcal{V}^*\right]$ and $\frac{\mathbb{E}\left[Y_{d_1} - Y_{d_2} \,\middle|\, \boldsymbol{V} \in \mathcal{V}^*\right]}{\mathbb{E}\left[Y_{d_3} - Y_{d_4} \,\middle|\, \boldsymbol{V} \in \mathcal{V}^*\right]}$,
   → **Sharp Bounds** and Parameter-inclusion Tests.

→ Fast, powerful inference via Chernozhukov, Lee, & Rosen (2013)

Estimation

1. **Semiparametric efficiency bound** for estimation of sharp bds.,
2. Formulate tractable "plug-in" estimator for sharp bounds,
3. **Cross-fitted bias-corrected estimator** for the bounds that:
   → Attains efficiency bound in suitably regular cases
   → Mitigates inward bias of "plug-in" estimator in irregular cases

IV Mixture Model: Basic Idea & Informal Outline

For any $d \in \mathcal{D}$ and each $z \in \mathcal{Z}$,

We observe: Y for those assigned $Z = z$ and took treatment d
We want to learn about: Y_d for those of type $\boldsymbol{T} = \boldsymbol{\tau}$, for $\tau \in \mathcal{T}$

Key: Individuals who take treatment $D = d$ when $Z = z$ consists of the mixture of individuals of types $\boldsymbol{T} = \boldsymbol{\tau}$ such that $\boldsymbol{\tau}(z) = d$.

→ Observed outcome distributions are mixture distributions:

$$\sum_{\boldsymbol{\tau} \in \mathcal{T}} \underbrace{\mathbb{1}\{\boldsymbol{\tau}(z) = d\}}_{\text{known, from model}} \underbrace{\mathbb{P}_{Q_0}(\boldsymbol{T} = \boldsymbol{\tau})\, \mathbb{P}_{Q_0}(Y_d \leq y \mid \boldsymbol{T} = \boldsymbol{\tau})}_{\text{unobserved}}$$
$$= \underbrace{\mathbb{P}_P(D = d \mid Z = z)\, \mathbb{P}_P(Y \leq y \mid D = d, Z = z)}_{\text{observed, can be estimated from data}} ,$$

→ Fix d and vary Z to get linear system with known coefficients
→ **Note:** Mixture only involves marginal distribution of Y_d

Informal Outline of Identification Approach and Results

0. TSM S determines set $\boldsymbol{\mathcal{T}}$ of possible response-types, and coefficients $\mathbb{1}\{\boldsymbol{\tau}(z) = d\}$ of treatment d-specific linear systems, which are collected into response-matrices $\mathbf{R}_d$.
→ Let $\boldsymbol{\pi} := (\mathbb{P}(\boldsymbol{T} = \boldsymbol{\tau}))_{\boldsymbol{\tau}}$ denote the (unknown) dist. of $\boldsymbol{T}$.
→ **Result:** S can be **sharply** represented by restriction $\boldsymbol{\pi} \in \boldsymbol{\mathcal{R}_\pi}(\mathrm{S})$
→ Focus on case where $\boldsymbol{\mathcal{R}_\pi}(\mathrm{S})$ is convex

1. For each d, can write linear system in terms of densities:

$$\mathbf{R}_d \boldsymbol{q}_d^Q(y) = \boldsymbol{p}_d^P(y) \quad \text{and} \quad \boldsymbol{q}_d(y) \geq 0$$

→ $\boldsymbol{p}_d$: observed densities $\boldsymbol{p}_{d|Z=z} = \mathbb{P}_P(D = d|Z = z) f_{Y|D=d,Z=z}$
→ $\boldsymbol{q}_d$: unobserved type-conditional densities $\mathbb{P}_{Q_0}(\boldsymbol{T} = \boldsymbol{\tau})\, f_{Y_d|\boldsymbol{T}=\boldsymbol{\tau}}$
→ **Result:** Id. set for Q_0 is set of Q such that $Q_Z = P_Z$ and

$$\text{(i) } \mathbf{R}_d \boldsymbol{q}_d^Q(y) \leq \boldsymbol{p}_d(y) \ \forall y \in \mathcal{Y} \,, \qquad (\star)$$
$$\text{(ii) } \boldsymbol{\pi}^Q \in \boldsymbol{\mathcal{R}_\pi}(\mathrm{S}) \,.$$

→ Relaxing to "$\leq$" is lossless when (ii) is imposed (note: $\int_{\mathcal{Y}} \boldsymbol{q}_d = \boldsymbol{\pi}$)

2. Consider set produced by integrating solutions to $(\star)$:

$$\Pi_d := \left\{ \int_{\mathcal{Y}} \boldsymbol{q}_d(y)\, dy \,\middle|\, \text{mble. } \boldsymbol{q}_d : \mathcal{Y} \to \mathbb{R}^{\mathcal{Z}}_{\geq 0} : \mathbf{R}_d \boldsymbol{q}_d \leq \boldsymbol{p}_d \right\} ,$$

→ Π_d is "Minkowski integral" of $y \mapsto \{\boldsymbol{x} \geq 0 \,|\, \mathbf{R}_d \boldsymbol{x} \leq \boldsymbol{p}_d(y)\}$
(similar to "Aumann Expectation" from Random Set Theory)
→ Π_d support function is integral of pointwise support functions

$$h_d(\boldsymbol{c}; \boldsymbol{p}_d(y)) := \sup_{\substack{\boldsymbol{x} \geq 0: \\ \mathbf{R}_d \boldsymbol{x} \leq \boldsymbol{p}_d(y)}} \boldsymbol{c}^\top \boldsymbol{x} = \inf_{\substack{\boldsymbol{\ell} \geq 0: \\ \mathbf{R}_d^\top \boldsymbol{\ell} \geq \boldsymbol{c}}} \boldsymbol{\ell}^\top \boldsymbol{p}_d(y) =: \boldsymbol{\ell}^*(\boldsymbol{c}; y)^\top \boldsymbol{p}_d(y)$$

→ **Key:** $\int_{\mathcal{Y}} h_d(\boldsymbol{c}; \boldsymbol{p}_d) = \sum_z \mathbb{E}_P\left[\boldsymbol{\ell}^*_{(z)}(\boldsymbol{c}; Y) \mathbb{1}\{D = d\} \,\middle|\, Z = z\right]$
→ **Main Result:** Id. set for $\boldsymbol{\pi}$ is simply the finite-dim. intersection

$$\Pi_{\mathrm{S}} := \boldsymbol{\mathcal{R}_\pi} \cap \bigcap_{d \in \mathcal{D}} \Pi_d$$

3. Supp. function of this convex intersection is the convex problem:

$$\inf_{\boldsymbol{u}} \left\{ \sum_{d \in \mathcal{D}} \sum_{z \in \mathcal{Z}} \mathbb{E}_P\left[\boldsymbol{\ell}^*_{(z)}(\boldsymbol{u}_d; Y) \mathbb{1}\{D = d\} \,\middle|\, Z = z\right] + h_{\boldsymbol{\mathcal{R}_\pi}(\mathrm{S})}\left(\boldsymbol{c} - \sum_{d \in \mathcal{D}} \boldsymbol{u}_d\right) \right\}$$

→ Moment ineq. problem, with P-dependent moment functions
→ Mean outcomes follow by generalizing to y-dependent $\boldsymbol{c}$, and enforcing (ii) on the set of means generated by sols. to $(\star)$.
→ **Main Result:** Each mean bound is **unique root** of such a problem.

Why Is This Tractable?

Computing $y \mapsto \ell^*(\boldsymbol{c}; y)$ requires solving a continuum of linear progs. (LPs).
→ These LP sol's. have the form $\left(\boldsymbol{c}^\top \mathbf{R}_{d;B}^{-1}\right)^\top$ for basis B and corresponding invertible submatrix $\mathbf{R}_{d;B}$ of $\mathbf{R}_d$
→ Given y, can find $B^*(y)$ that is optimal for $(\boldsymbol{c}; \boldsymbol{p}_d(y))$ by solving "small" LP
→ **Key:** Once $B^*(y)$ found, easily find y' where $B^*(y)$ **remains optimal** and set $B^*(y') := B^*(y)$
→ Can find $y \mapsto B^*(y)$ over full Y support in **finitely many steps**

Fix $d \in \mathcal{D}$. For $y \in \mathcal{Y}$, the set of solutions to $(\star)$ defines a convex polytope,

$$\mathcal{B}_d(\boldsymbol{p}_d(y)) := \{\mathbf{x} \geq 0 \mid \mathbf{R}_d \mathbf{x} \leq \boldsymbol{p}_d(y)\} \,.$$

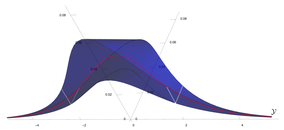
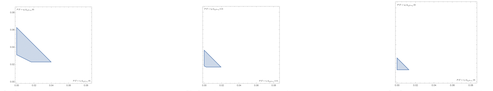


Figure 1: Hypothetical graph of the correspondence $y \mapsto \mathcal{B}_d(\boldsymbol{p}_d(y))$



(a) $\mathcal{B}_d(\boldsymbol{p}_d(0))$ **(b)** $\mathcal{B}_d(\boldsymbol{p}_d(1.5))$ **(c)** $\mathcal{B}_d(\boldsymbol{p}_d(2))$

Figure 2: In this example, (a) and (b) can be solved using just 1 LP, (c) requires a new LP. In fact, **entire continuum here solved via 2 LPs**: one for $y \in (-2, 2)$ and one for tails.

Estimating the Sharp Bounds

Consider ratio parameter $\theta := \mathbb{E}[Y_d \,|\, \boldsymbol{T} \in A] =: \boldsymbol{c}_A^\top \boldsymbol{\eta} / \boldsymbol{c}_A^\top \boldsymbol{\pi}$.

Result, in paper: Upper bound $\bar{\theta}(P)$ defined by estimation equation

$$\overline{\Psi}(\bar{\theta}(P); P) = 0 \,,$$

where $\gamma \mapsto \overline{\Psi}(\gamma; P)$ is a decreasing function of the form

$$\overline{\Psi}(\gamma; P) := \inf_{\boldsymbol{u} \in <\text{known subset of } \mathbb{R}^{\mathcal{Y}}>} \sum_{z \in \mathcal{Z}} \mathbb{E}_P\left[\boldsymbol{m}^P_\theta(\gamma, \boldsymbol{u}; Y, D, Z) \,\middle|\, Z = z\right] ,$$

can solve for root via convex programming, if P is known.

- In practice, don't know P, so can "plug-in" estimate $\tilde{P}_n$
- $\tilde{P}_n$ usually smoothed and projected estimate to ensure non-empty identified set → can cause regularization bias
- Also, infimum of estimated objective $\implies \mathbb{E}_P\left[\overline{\Psi}\left(\gamma; \tilde{P}_n\right)\right] < \overline{\Psi}(\gamma; P)$
  → can cause "overfitting" bias, as follows:

$$\overline{\Psi}\left(\bar{\theta}\left(\tilde{P}_n\right); \tilde{P}\right) < \overline{\Psi}\left(\bar{\theta}\left(\tilde{P}_n\right); P\right)$$
$$\implies \tilde{\bar{\theta}}_n := \bar{\theta}\left(\tilde{P}_n\right) < \bar{\theta}(P), \text{ since } \overline{\Psi} \downarrow \rightarrow \text{overfitting bias is inward}$$

Bias-Corrected Estimator. Under sufficient regularity, have linearization:

$$0 = \bar{\Psi}\left(\bar{\theta}(P); P\right)$$
$$\approx \bar{\Psi}\left(\tilde{\bar{\theta}}_n; \tilde{P}\right) + \sum_{z \in \mathcal{Z}} \left(P_{Z=z} - \tilde{P}_{Z=z}\right) \boldsymbol{m}^{\tilde{P}}_\theta\left(\tilde{\bar{\theta}}_n, \tilde{\boldsymbol{u}}, \tilde{\boldsymbol{\lambda}}_n; \cdot\right) - \left(\boldsymbol{c}_A^\top \tilde{\boldsymbol{\pi}}\right)\left(\bar{\theta}(P) - \tilde{\bar{\theta}}_n\right)$$

Suggests corrected estimator:

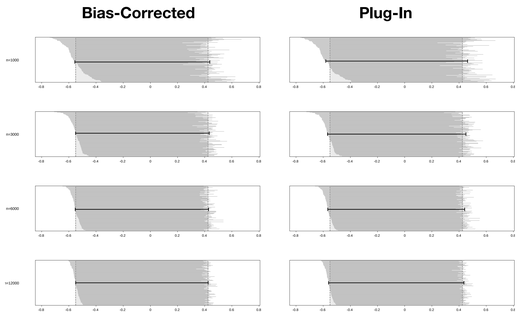
$$\bar{\theta}(P) \approx \tilde{\bar{\theta}}_n + \frac{1}{\boldsymbol{c}_A^\top \tilde{\boldsymbol{\pi}}} \sum_{z \in \mathcal{Z}} \mathbb{E}_P\left[\boldsymbol{m}^{\tilde{P}}_\theta\left(\tilde{\bar{\theta}}_n, \tilde{\boldsymbol{u}}, \tilde{\boldsymbol{\lambda}}; Y, D, Z\right) \,\middle|\, Z = z\right] ,$$

→ To address overfitting: cross-fit, estimate correction on held-out fold.
→ When sufficiently regular, bias-corrected estimator is efficient.
→ **Key:** When irregular, correction renders $\sqrt{n}$-inward-error second-order via primal-dual inequalities!

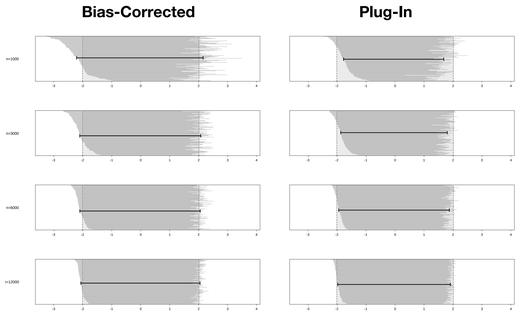
Simulations

Estimation

Regular Case



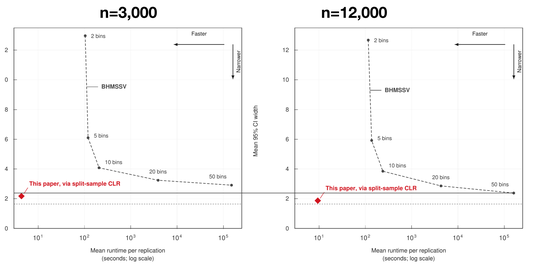
Irregular Case



Inference

Comparison vs. Discretization Approach

Runtime vs. Mean 95% CI Width



(DGP from Figure 1 of Bai, Huang, Moon, Santos, Shaikh, & Vytlacil (2026))

Power Curves

